HIGHER SECONDARY - SECOND YEAR - MAR - 2024

KEY ANSWERS FOR PHYSICS

PART - I

1. a) Photovoltaic action
2. c) $900 Vm^{-1}$
3. c) 480 W
4. a) 3
5. c) Polarisation
6. a) $Q/\sqrt{2}$
7. d) $\frac{3}{\pi} P_m$
8. d) its Wavelength
9. b) $\pi/4$
10. a) more than before
11. d) 6 μF
12. d) 3750 A°
13. a) Plane Polarasid
14. a) Albert Einstein
15. c) 1.1 eV

11) ஒளி மின்னிலை விளைவினால்
12) $900 Vm^{-1}$
13) 480 W
14) 3
15) நிலைமமாக்கல்
16) $Q/\sqrt{2}$
17) $\frac{3}{\pi} P_m$
18) அதன் அலைநீளம்
19) $\pi/4$
20) முன்னை விட அதிகமாக இருக்கும்
21) 6 MF
22) 3750 A°
23) தளவாக நிலைமமாக்கல்
24) ஆல்பர்ட் ஐன்ஸ்டீன்
25) 1.1 eV

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PART-II

Qn No. 24 is Compulsory:

16. **Hysteresis:**
Lagging of Magnetic induction behind the magnetising field.

17. **Malus' Law:-** When a beam of plane Polarised light of intensity I_0 is incident on an analyser the intensity of light I transmitted from the analyer varies directly as the square of cosine of angle θ b/w the transmission axes of Polariser & analyser.

$$I = I_0 \cos^2\theta$$

18. **Electrostatic Potential:** Work done by an external force to bring a unit +ve charge with Cnst. Velocity from infinity to that Point.

$$V = \frac{q}{4\pi\varepsilon_r}$$

19. $\varepsilon = \frac{d\phi}{dt} = \frac{4\times10^{-3}}{0.4} = 10\times10^{-3}\,V$ (or) 10mV.

20. **Application of Seebeck Effect:**
i) Thermoelectric generators ii) Automobiles
iii) Thermocouples & Thermopiles.

21. $\lambda = \frac{0.6931}{T_{1/2}} = \frac{0.6931}{5.01\times24\times60\times60} = 1.6\times10^{-6}\,s^{-1}$

22. **Electromagnetic Wave:-** its non-mechanical wave which moves with speed equal to speed of light.

23) **Biasing:-** It Provides external energy to charge carriers to overcome the barrier Potential & made them move in Particular direction. **Types:** i) Forward (ii) Reverse

24.) $P = \frac{1}{f} = \frac{1}{1.5\,m} = 0.67\,D$

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PART-III

Qn: No: 33 is Compulsory:

25. Atomic No:- decreased by one & its mass number remains Same.

$$_{Z}X^{A} \rightarrow {}_{Z-1}Y^{A} + e^{+} + \nu$$

$$P \rightarrow n + e^{+} + \nu$$

$$_{11}Na^{22} \rightarrow {}_{10}Ne^{22} + e^{+} + \nu$$

26. $V_d = \frac{I}{neA} = \frac{0.2}{8.4\times10^{28}\times1.6\times10^{-19}\times0.5\times10^{-6}} = 0.03\times10^{3}\,ms^{-1}$

27. Lenses in Contact:-

Diagram

Explanation

$\frac{1}{v'} - \frac{1}{u} = \frac{1}{f}$; $\frac{1}{v} - \frac{1}{v'} = \frac{1}{f_2}$; $\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}}$

28. Current Sensitivity:- Deflection Produced for unit Current

i) increasing No. of turns ii) increasing B (iii) increasing Area

iv) decreasing Small Couple Per unit twist.

29. $n = \frac{Pt}{hc} = \frac{50\times10^{-3}\times64\times10^{9}}{6.626\times10^{-34}\times3\times10^{8}} = 1.61\times10^{17}\,s^{-1}$

30. Diagram

Explanation

$B = \mu_0 n i$; $\phi_B = \mu_0 n i A$; $N\phi_B = \mu_0 n^2 A i l$; $N\phi_B = Li$

$\boxed{L = \mu_0 n^2 A l}$

31. Any 3 difference.

32. Diagram, Explanation.

$\phi_E = \oint \vec{E}\cdot d\vec{A} = \oint E\,dA\cos\theta = \oint E\,dA$

$\phi_E = E\cdot 4\pi r^2 = \frac{Q}{4\pi\varepsilon_0 r^2}\cdot 4\pi r^2$; $\boxed{\phi_E = Q/\varepsilon_0}$

33. $\lambda = \frac{hc}{E_g} = \frac{6.6\times10^{-34}\times3\times10^{8}}{1.875\times1.6\times10^{-19}} = 660\,nm$

$\boxed{\text{RED Colour}}$

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PART - IV

34) a) **Simple Microscope:-** Explanation

Near Pt:- focussing:- Diagram, $m = \frac{v}{u} = \frac{D}{u}$

$m = \frac{v}{u} = 1 - v/f = 1 + D/f$

Normal focussing:- Diagram, $m = \theta_i/\theta_o = \frac{h/f}{h/D}$

$m = D/f$

b) **Metre Bridge:-** Diagram Explanation

$\frac{P}{Q} = \frac{R}{S} = \frac{rAJ}{r.JB} = l_1/l_2$; $P = Q \cdot l_1/l_2$

End Resistance:- $P = \rho l/A$; $\rho = \frac{PA}{l} = \frac{P\pi r^2}{l}$

35) a) **Circular Coil:-** Diagram, Explanation

Upto $dB = \frac{\mu_0}{4\pi}\frac{i\,dl}{r^2}$; Upto $\vec{B} = \frac{\mu_0 N i}{2}\frac{R^2}{(R^2+z^2)^{3/2}}\hat{k}$

$$\boxed{\vec{B} = \frac{\mu_0 N i}{2}\hat{k}}$$

b) **Angle of deviation:-** Diagram, Explanation

Upto $d = (i_1 + i_2) - (r_1 + r_2)$

$d = i_1 + i_2 - A$

$\eta = \frac{\sin(A+D/2)}{\sin A/2}$

36) a) **Einsteins Photoelectric Equation:-** Diagram, Explanation

$h\nu = \phi_0 + \frac{1}{2}mv^2$; $h\nu_0 = \phi_0$; $h\nu = h\nu_0 + \frac{1}{2}mv^2$; $K_{max} = \frac{1}{2}mv_m^2$

$K_{max} = h\nu - \phi_0$

b) **Pure inductor:-** Circuit diagram, Explanation

$V = V_m \sin\omega t$; $\varepsilon = -L(di/dt)$; $i = (V_m/L)\int \sin\omega t$

$i = (V_m/L\omega)(-\cos\omega t)$; $i = i_m \sin(\omega t - \pi/2)$

$\therefore$ Current lags behind the Voltage by $\pi/2$

Phasor diagram, Wave diagram.

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37) a) FM:- 3 - Advantages
3 - Limitations

b) Maxwell's Modifications of Ampere's Circuital law:-
$\oint \vec{B}\cdot d\vec{l} = \mu_0 i_c$; $\oint \vec{B}\cdot d\vec{l} = 0$; $i_d = \varepsilon_0 \frac{d\phi_E}{dt}$
$\oint \vec{B}\cdot d\vec{l} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$; Diagram

38) a) Axial line:- Diagram, Explanation
$\vec{E}_+ = \frac{q}{4\pi\varepsilon_0 (r-a)^2}\hat{p}$; $\vec{E}_- = -\frac{q}{4\pi\varepsilon_0 (r+a)^2}\hat{p}$; $\vec{E} = \vec{E}_+ + \vec{E}_-$
$\boxed{P = 2qa\hat{p}}$ $\vec{E} = \frac{2\vec{p}}{4\pi\varepsilon_0 r^3}$

b) Nuclear Reactor:- Nuclear Reaction takes place in Self-Sustained Controlled manner & energy produced is used either for research purpose for Power generation
i) Moderator:- Explanation with Example
ii) Control rod:- Explanation with Example
iii) Cooling System:- Explanation with Example.

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